# Machine Learning Foundations: Geometry, Optimization, and Linear Classification

From vectors and planes to loss functions, feature vectors, and classifiers

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**Point / Vector**

$x=(3,2,2)$

**Plane / Separator**

$3x_1+x_2-1=0$

**Loss / Optimization**

$\theta \leftarrow \theta-\gamma\nabla L(\theta)$

**Classifier**

$h(x)\in\{-1,+1\}$

Geometry gives us representation

Optimization gives us learning

Classification gives us prediction

Foundations for early machine learning concepts and linear classification

# Part I — Framing, Notation, and Roadmap

How the course’s mathematical language becomes machine learning language

vector hyperplane classifier

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# Why These Foundations Come First

line / plane → classifier boundary

**Linear Algebra / Geometry**

Gaussian curve → uncertainty

**Probability / Uncertainty**

**Machine Learning**

**Calculus / Optimization**

bowl surface → optimization

**Computation / NumPy**

movie → feature vector

## Core idea

Machine learning is built from representation, geometry, optimization, uncertainty, and computation.

## Why these prerequisites matter

- Vectors represent examples and parameters
- Planes represent decision boundaries
- Loss functions define prediction error
- Gradients tell us how to improve
- Probability helps us model uncertainty
- Computation makes the math executable

Before a model can learn, it needs representation, error measurement, and a method for improvement.

# The Big Story: From Data to Decisions

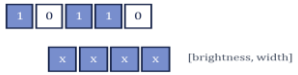
Machine learning turns real-world objects into numerical representations, places them in a geometric space, and learns rules that predict labels or values.

### Raw Object

movie
handwritten digit

raw input

### Feature Vector




encoded measurable properties

$x \in R^d$

### Point in Space

vector becomes a location in geometry

point / vector

### Model / Boundary

separate or score regions of the space

$\theta \cdot x + \theta_0$

### Prediction / Error

predict, compare, measure error

$h(x), L(\theta)$

**Running examples**

movie → [1,0,1,1,0] → like / dislike

digit → [brightness, width] → class label

Takeaway: machine learning first turns objects into vectors, then learns decision rules in geometric space.